

Logic in AI

Università degli Studi di Verona
Imbriani Paolo, VR553697
paolo.imbriani@studenti.univr.it
Professor Margherita Zorzi

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1 Introduction

There are two main goals for this course:

- Introducing a notion of deduction
- Introducing Curry Howard Isomorphism, that connect the notion of computation with the notion of deduction. A proof is just a mathematical object that can be manipulated and transformed.

Definition: Logical System

A logical system has at least three components:

- Syntax (an alphabet, a grammar, a set of well-formed formulas)
- Semantics (in order to link a meaning to the syntax)
- Deductive system (a set of rules to derive new formulas from the existing ones)

Perhaps, in the past you have encountered a logical system like Natural Deduction, tableaux, or sequent calculus.

1.1 Areas of interest

There are several applications of logic in AI, took directly from SOTA (state of art) papers:

- **"Theoritical" Computer Science**: automata, formal languages, computability or recursion theory, complexity theory, etc.
- **Artificial Intelligence**: deduction systems (proof theory), expert systems, NLP, automated proof, multi-agent systems, etc.
- **Programming Languages**: logical programming, resolution functional programming, semantics, type systems, etc.
- **Software Engineering**: program verification, program design, speficications, etc.
- **Hardware**: circuit design verification, etc.

1.2 Roadmap

We will be following a roadmap that is based on the following topics:

1. Propositional logic
   - Natural deduction, with which we will use it to reason about logical principles.
2. First Order Logic
3. Intuitionistic logic
   - $\underbrace{\vdash \varphi \Longleftrightarrow \vDash \varphi}_{\leftarrow}$
   - λ calculus
   - Curry Howard
4. Modal Logics $\boxed{?}, \Diamond$

2 Propositional Logic

We are going to start with propositional logic, which is the simplest logical system.

2.1 Syntax

Propositional logic is composed by:

- Proposition symbols: AT $= \{p_0, p_1, p_2, ...,\} \cup \bot$ ($|\mathbb{N}| = \aleph_0$)
- Connectives:
  - Negation: $\neg\varphi$
  - Conjunction: $\varphi \wedge \psi$
  - Disjunction: $\varphi \vee \psi$
  - Implication: $\varphi \to \psi$
  - Bi-implication: $\varphi \leftrightarrow \psi$
- Auxialiary symbol: $(,)$

The set PROP of propositions is the *smallest* set X with the properties:

1. $p_i \in X (i \in \mathbb{N}), \bot \in X$
2. $\varphi, \psi \in X \Rightarrow (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \to \psi), (\psi \leftrightarrow \varphi) \in X$
3. $\varphi \in X \Rightarrow (\neg\varphi) \in X$

We need the following induction principles in order to use proof by induction:

Theorem: Induction Principle

Let A be property when $A(\varphi)$ holds for all $\varphi \in$ PROP if:
1. $A(p_i)$ holds for all $i \in \mathbb{N}$ and $A(\bot)$ holds
2. $A(\varphi)$ and $A(\psi)$ hold implies $A(\varphi \wedge \psi)$, $A(\varphi \vee \psi)$, $A(\varphi \to \psi)$, $A(\psi \leftrightarrow \varphi)$ hold
3. $A(\varphi)$ holds implies $A(\neg\varphi)$ holds

Theorem: Definition by Recursion

Let mappings $H_{\boxed{?}} : A^2 \to A$ and $H_\neg : A \to A$ be given and let H_{at} be a mapping from the set of atoms into A, then there exists exactly one mapping $F :$ PROP $\to A$ such that

Theorem: Primitive Recursion

$\exists! f : \mathbb{N} \to \mathbb{N}$ and $h : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ where $c \in \mathbb{N}$ such that $f(0) = c$ and $\forall n \in \mathbb{N}, f(n+1) = H(f(n))$:

- Ω class of satisfying sets Z
  - $(0, c) \in Z, (n, x) \in Z \Rightarrow (s(n), h(n, x))$
  - f intersection of all the sets above
  - f is a set and in Ω

We prove by induction that $\forall n \in \mathbb{N}, \exists! a \in \mathbb{N}$ such that $s \vdash (n, a) \in f$.

- **Base case**: $(0, c) \in f$ by definition. Suppose $\exists d \neq c$ and $f \setminus \{(0, d)\}$ is satisfied and properly contained in f.

Since I am violating the fact that f is the smallest one, this is impossible.

- **Inductive case**: Suppose $\exists! w (i, w) \in f$. By construction of f:

$$(s(i), h(i, w)) \in f$$

suppose $c \neq h(i, w)$ s.t. $(s(i), e) \in f$.

$$f \setminus \{(s(i), e)\} \text{ is satisfied and properly contained in } f$$

$$\Rightarrow \text{impossible}$$

2.2 Semantics

The following two definitions are equal:

Definition: 1

A mapping $v : \text{PROP} \to \{0, 1\}$ is a valuation if:

- $v(p_i) \in \{0, 1\}$ for all $i \in \mathbb{N}$ and $v(\bot) = 0$
- $v(\varphi \wedge \psi) = \min(v(\varphi), v(\psi))$
- $v(\varphi \vee \psi) = \max(v(\varphi), v(\psi))$
- $v(\varphi \to \psi) = \max(1 - v(\varphi), v(\psi))$
- $v(\varphi \leftrightarrow \psi) = 1 \Leftrightarrow v(\varphi) = v(\psi)$
- $v(\neg\varphi) = 1 - v(\varphi)$

Definition: 2

A mapping $v : \text{PROP} \to \{0, 1\}$ is a valuation if:

- $v(p_i) \in \{0, 1\}$ for all $i \in \mathbb{N}$ and $v(\bot) = 0$
- $v(\varphi \wedge \psi) = 1$ iff $v(\varphi) = v(\psi) = 1$
- $v(\varphi \vee \psi) = 1$ iff $v(\varphi) = 1$ or $v(\psi) = 1$
- $v(\varphi \to \psi) = 0$ iff $v(\varphi) = 1$ and $v(\psi) = 0$
- $v(\varphi \leftrightarrow \psi) = 1$ iff $v(\varphi) = v(\psi)$
- $v(\neg\varphi) = 1$ iff $v(\varphi) = 0$

2.3 Tautology

Definition: 3

Let Γ be a set propositions, φ is a tautology if $\forall v$ valuation,

$$\Gamma \vDash \Leftrightarrow \forall v : ([\psi]_v = 1 \quad \text{for all} \quad \psi \in \Gamma) \Leftrightarrow [\varphi]_v = 1$$

2.3.1 Reductio ad absurdum

What does it mean having the bottom as a logical consequence? We have:

$$\Gamma, \neg\alpha \vDash \bot \Rightarrow \Gamma \vDash \alpha$$

Let's provide a valuation for the bottom symbol.

$$\forall v.[|\bot|]_v = 0$$

But if I have something like this:

$$\Delta \vDash \bot$$

Let's try to reason about it using the definition:

$$\forall v.[|\Delta|]_v = 1 \Rightarrow [|\bot|]_v = 1$$

Now let's see the meta-conjuction as a disjunction using this theorem:

$$A \to B := \neg A \vee B$$

So:

$$\underbrace{[|\Delta|]_v \neq 1}_{\forall v.\exists\gamma\in\Delta \text{ s.t. } [|\gamma|]_v=0} \vee \underbrace{[|\bot|]_v = 1}_{\text{false}}$$

This is the meaning of having the bottom has a logical consequence. So:

$$\forall v.\exists\gamma \in \Delta \text{ s.t. } [|\gamma|]_v = 0 \to \Gamma \cup \{\neg\alpha\} \text{ unsat so } \Gamma \vDash \alpha$$

From this, we get the following principles:

- $\neg\alpha \iff \alpha \to \bot$
- $\Gamma, \alpha \vDash \beta \iff \Gamma \vDash \alpha \to \beta$

Let's unwrap this using the rules we have just shown:

$$\Gamma, \neg\alpha \vDash \bot \Longrightarrow \neg\alpha \to \bot$$

So we can derive intuitively the following:

$$\neg\neg\alpha \equiv \alpha$$

We proved the left ($\leftarrow$) direction, but let's now prove the right ($\rightarrow$) direction:

$$\forall v.([|\Gamma|]_v = 1 \Rightarrow [|\alpha|]_v = 1) \iff \forall v.([|\Gamma|]_v = 1 \Rightarrow [|\neg\neg\alpha|]_v = 1)$$

2.4 Natural Deduction

We now consider the persective of the so-called: inference making. The origin of modern notion of inference making? **David Hilbert** proposed a system called **Hilbert-style Deductive System** which is a set of axioms and inference rules. Several mathimaticians understood that this style is not very close to the usual practice of reasoning, so Gentzen proposed **Sequent Calculi** and **Natural Deduction**. In the 60′, Dag Brawitz structured Natural Deduction and wrote a beutiful reference called **Natural Deduction: A Proof Theoritical Study**, which is the very first step towards automated theorem proving.

The structure of Natural Deduction is based on the notion of a *proof tree*, which is a tree where each node is a formula and each edge is an inference rule. The root of the tree is the conclusion and the leaves are the premises.

$$\text{Label}\ \frac{\text{Premise 1} \qquad \text{Premise 2} \qquad \text{Premise 3}}{\text{Conclusion}}\ \text{Rule name}$$

There are usually two kind of rules:

- **Introduction rules**: they introduce a new connective in the conclusion.
- **Elimination rules**: they eliminate a connective from the premises.

For example:

- Elimination rule for implication:

$$\frac{\varphi \to \psi \qquad \varphi}{\psi} \to E$$

- Introduction rule for implication:

$$\frac{\Gamma, \varphi \vdash \psi}{\Gamma \vdash \varphi \to \psi} \to I$$

Let's prove some basic tautologies:

- $A \to A$

$$\frac{[A]^1}{A \to A} \to I^1$$

Let's now add the **Conjunction** rule:

- Introduction rule for conjunction:

$$\frac{\varphi \qquad \psi}{\varphi \wedge \psi} \wedge I$$

- Elimination and introduction rule for conjunction:

$$\frac{\varphi \wedge \psi}{\varphi} \wedge E_1$$

$$\frac{\varphi \wedge \psi}{\psi} \wedge E_2$$

- Introduction and elimination rule for disjunction:

$$\frac{\varphi}{\varphi \vee \psi} \vee I_1$$

$$\frac{\psi}{\varphi \vee \psi} \vee I_2$$

$$\frac{\varphi \vee \psi \qquad \Gamma, \varphi \vdash \chi \qquad \Gamma, \psi \vdash \chi}{\Gamma \vdash \chi} \vee E$$

  Elimination rule for disjunction is a bit more complex, since it requires two premises and a conclusion and it introduce proof by cases.
- Ex falso:

$$\frac{\bot}{\varphi} \bot E$$

  Ex falso is accepted also in intuitionistic logic, but the following is not:
- Reductio ad Absurdum (RAA), a **strong** classical principle

$$\frac{\Gamma, \neg\varphi \vdash \bot}{\Gamma \vdash \varphi}\,\text{RAA}$$

- Tertium non datur:

$$\frac{}{\varphi \vee \neg\varphi}\,\text{TND}$$

  In a classical context you always know if a formula is true or false. In order to prove this, I need to use the RAA rule.

$$\dfrac{\dfrac{\neg\alpha \vee \alpha \qquad \neg(\alpha \vee \neg\alpha)}{\bot}\to E}{\alpha \vee \neg\alpha}$$

Theorem: Pierce's law

Pierce's law is a theorem in propositional logic that states:

$$((P \to Q) \to P) \to P$$

The computational equivalent of Pierce's law is the type of the constructor **call_cc** in Scheme. The derivation of Pierce's law is interesting since we need to use the classical principles:

$$\dfrac{\dfrac{\dfrac{\dfrac{[\alpha]^1 \qquad [\neg\alpha]^2}{\bot}\to E}{\beta}\text{EF}}{\alpha \to \beta}\to I^1 \qquad [(\alpha \to \beta) \to \alpha]^3}{\alpha}\to E \qquad \neg\alpha}{\dfrac{\dfrac{\bot}{\alpha}\text{RAA}^2}{((\alpha \to \beta) \to \alpha) \to \alpha}\to I^3}\to E$$

This is the derivation of the Pierce's law. Redundant rules can be eliminated, in some other deductive systems, but in Natural Deduction we need to keep them so we can close the derivation tree.

2.4.1 Derivation

A derivation φ is a finite tree of formulas where each node is a formula and each edge is an inference rule. The root of the tree is the conclusion and the leaves are the premises.

Definition: Derivability

Let $\mathcal{L}$ be a language and Γ a set of formulas

1. $\Gamma \subseteq \text{WFF}_{\mathcal{L}}$ derives φ ($\Gamma \vdash \varphi$) if there exists $\frac{\Pi}{\varphi}$ s.t. $\mathcal{H}p[\Pi] \subseteq \Gamma$
2. φ is a theorem if $\varnothing \vdash \varphi$

Definition: Eliminability

An inference rule R is eliminable if for each derivation $\frac{\Pi}{\varphi}$ in which R is used there exists another derivation $\frac{\Pi^*}{\varphi}$ in which R is not used and $\mathcal{H}p[\Pi^*] \subseteq \mathcal{H}p[\Pi] \subseteq \Gamma$.

Definition: Equivalence

Let's take two different deductive systems I and J. I and J are equivalent if:

$$\forall \Gamma, \varphi, \Gamma \underset{I}{\vdash} \varphi \iff \Gamma \underset{J}{\vdash} \varphi$$

Exercise for home: Prove that we can sussume / replace in the derivation, EX FALSO by RAA and in the hint is copying the following proof:

Theorem: Example

Let's define $\mathcal{N}$ natural deduction, with this set of rules: {EF, RAA, TND, $\to I/E$, $\wedge I/E$, $\vee I/E$}.

$$\mathcal{N}^* := \mathcal{N} \setminus \{\text{RAA}\} \cup \{\text{TND}\}$$

and $\mathcal{N}$ and $\mathcal{N}^*$ are equivalent.

Proof: $\forall \varphi \ \frac{\Pi}{\varphi}$ in $\mathcal{N}, \exists \frac{\Pi^*}{\varphi}$ in $\mathcal{N}^*$

$$\mathcal{H}p[\Pi] = \mathcal{H}p[\Pi^*]$$

1. Base case: $\Pi \equiv \varphi$ is a derivation of length 1, then $\Pi^* = \Pi$.
2. Inductive step: $\Pi = \frac{\Pi_1 \dots \Pi_k}{\varphi} r$
   1. $r \neq$ RAA by IH

$$\frac{\Pi_1^* \dots \Pi_k^*}{\varphi} r = \Pi^*$$

   2. $r =$ RAA

$$\Pi = \frac{\dfrac{\Pi_1}{\varphi}\,\text{RAA}}{\varphi}$$

$$\Pi^* = \frac{\dfrac{\Pi_1^*}{\varphi}\,\text{TND}}{\varphi}$$

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2.5 Soundness and Completeness

There are two main property about a logic that we would like to connect: syntax and semantics. To do that, we want to have soundness and completeness for the System.

$$\vdash \varphi \iff \models \varphi$$

Where $\implies$ is **soundness** and $\impliedby$ is **completeness**.

There's a nice application of the soundness theorem which is the following one:

Note:

$$\vdash \varphi \Longrightarrow \vDash \varphi \sim \nvDash \varphi \Longrightarrow \nvdash \varphi$$

since

$$A \to B \Leftrightarrow \neg B \to A$$

3 First Order Logic

First Order Logic is a logical system that extends propositional logic by adding quantifiers and variables. It allows us to reason about objects and their properties.

Sometimes I would like to express a formal language. Let's take this language:

$$\mathcal{L} = \{a^n b^n \mid n \geq 0\}$$

We have an alphabet:

$$< r_1, ..., r_n; a_1, ..., a_m; k > \text{ with } r_i \geq 0, a_j > 0$$

- Predicate symbols: $P_1, ..., P_n$ where n is the arity of the predicate.
- Function symbols: $f_1, ..., f_m$ where m is the arity of the function.
- Constant symbols: $c_1, ..., c_k$ where k is the number of constants.
- Variables: $x_1, ..., x_l$ where l is the number of variables. (countably many: $|\text{VAR}| = \aleph_0$)
- Connectives: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow, \bot, \forall, \exists$
- Auxialiary symbols: $(,)$

Definition: Terms

TERM is the smallest set X with the properties:

- $c_i \in X (i \in I)$ and $x_i \in X (i \in N)$.
- $t_1, ..., t_{a_i} \in X \Longrightarrow f_{i(t_1,...,t_{a_i})} \in X$ for $1 \leq i \leq m$.

TERM is our *set of terms.*

Definition: Formulas

FORM is the smalles set X with the properties:

- $\bot \in X; P_i \in X$ if $r_i = 0; t_1, ..., t_{r_i} \in \text{TERM} \Longrightarrow P_{i(t_1,...,t_{r_i})} \in X; t_1, t_2 \in \text{TERM} \Rightarrow t_1 = t_2 \in X$,
- $\varphi, \psi \in X \Longrightarrow (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\psi \leftrightarrow \varphi) \in X$,
- $\varphi \in X \Longrightarrow (\neg\varphi) \in X$,
- $\varphi \in X \Longrightarrow (\forall x_i \varphi), (\exists x_i \varphi) \in X$.

Definition: Induction for terms

Let $A(t)$ be a property of terms. If it holds for t a variable or a constant and if $A(t_1), ...A(t_n) \Rightarrow A(f(t_1, ..., t_n))$ for all function f then $A(t)$ holds for all $t \in \text{TERM}$.

Definition: Induction for formulas

Let $A(\varphi)$ be a property of formulas. If:

- $A(\varphi)$ for atomic φ
- $A(\varphi)$ and $A(\psi)$ implies $A(\varphi \wedge \psi), A(\varphi \vee \psi), A(\varphi \rightarrow \psi), A(\psi \leftrightarrow \varphi)$
- $A(\varphi)$ implies $A(\neg\varphi)$

- $A(\varphi)$ implies $A((\forall x_i)\varphi, A((\exists x_i)\varphi)$ for all i then $A(\varphi)$ holds for all $\varphi \in \text{FORM}$.

Definition: Free variables

The set $\text{FV}(t)$ of free variables of t is defined by:

- $\text{FV}(x_i) := \{x_i\}$, $\text{FV}(\overline{c}_i) := \varnothing$
- $\text{FV}\left(f_{i(t_1,\ldots,t_{a_i})}\right) := \text{FV}(t_1) \cup \ldots \cup \text{FV}(t_n)$

Definition: Free variables for formuals

The set $\text{FV}(t)$ of free variables of t is defined by:

- $\text{FV}(P(t_1, \ldots, t_n)) := \text{FV}(t_1) \cup \ldots \cup \text{FV}(t_n)$
- $\text{FV}(t_1 = t_2) := \text{FV}(t_1) \cup \text{FV}(t_2)$
- $\text{FV}(\varphi \wedge \psi) := \text{FV}(\varphi) \cup \text{FV}(\psi)$, $\text{FV}(\varphi \vee \psi) := \text{FV}(\varphi) \cup \text{FV}(\psi)$, $\text{FV}(\varphi \to \psi) := \text{FV}(\varphi) \cup \text{FV}(\psi)$, $\text{FV}(\psi \leftrightarrow \varphi) := \text{FV}(\varphi) \cup \text{FV}(\psi)$
- $\text{FV}(\forall x_i \varphi) := \text{FV}(\exists x_i \varphi) := \text{FV}(\varphi) - \{x_i\}$

Definition: Closed variables

φ is closed variable if

$$\text{FV}(\varphi) = \varnothing$$

A closed formula is also called a *sentence*. A formula without quantifiers is called *open* formula.

Definition: Subformulas

A subformula can be inductively defined:

- $\text{sub}(\varphi) = \{\varphi\}$
- $\text{sub}(\varphi \odot \psi) = \text{sub}(\varphi) \cup \text{sub}(\psi) \cup \{\varphi \odot \psi\}$ for $\odot \in \{\wedge, \vee, \to, \leftrightarrow\}$
- $\text{sub}(\neg\varphi) = \text{sub}(\varphi) \cup \{\neg\varphi\}$
- $\text{sub}(\forall x_i \varphi) = \text{sub}(\exists x_i \varphi) = \text{sub}(\varphi) \cup \{\forall x_i \varphi, \exists x_i \varphi\}$

Definition: Free and Bound occurrences

An occurence of a variable x in δ is BOUND, if x occurs in $\varphi \in \text{sub}\{\delta\}$ and $\varphi \equiv \{\text{Q}x.\varphi\}$ … FINIRE A CASA

Definition: Substitution

We define substitution of a variable x with a term t in a formula φ as follows:

$$\varphi[t/x]$$

if x is free:

$$\varphi[t/x] = (...t...)$$

if x is bound:

$$\varphi[t/x] = (...x...)$$

So t needs to be free for x in φ.

But first let's show some other theorem using Natural Deduction.

Example

Proof.

$$\vdash \underbrace{(\varphi \to \exists x.\varphi(x))}_{\alpha} \to \underbrace{\exists x.(\varphi \to \psi(x))}_{\beta}$$

This is not a theorem in general, we'll need to fix some constraints: $x \notin FV(\varphi)$. We could try to understand if we are dealing with classical principles. Let's break down how to approach this derivation:

- $(\varphi \to \exists x.\varphi(x))$ will probably need to be derived using the $\to I$ rule, so we need to assume φ and derive $\exists x.\varphi(x)$. So

$$(\varphi \to \exists x.\psi(x)) \in \mathcal{H}p[\Pi]$$

- I "destroy" these subformulas: $\varphi \to \exists x.\psi(x)$ since I want to extract the body formula ψ and use it inside the second part of the theorem. The idea is to use $\psi(x) \in \beta$.
- $\exists x.\psi(x)$ eliminate the $\exists$ and derive from $\psi(x)$ something of interest like β with $\exists E$.
- Try to start from $\neg\beta$ to derive $\bot$ and then derive β.)

$$\dfrac{\dfrac{\dfrac{

If for each structure it's satisfafiable then we can derive it. This proof was firstly formulated by Kurt Godel. Before jumping into the proof wen need to introduce an important concept of **Consistent Sets**. Note that WFF stands for the set of well-formed formulas.

$$\Gamma \subseteq WFF_{\mathcal{L}} \text{ is consistent if } \Gamma \nvdash \bot$$
$$\text{not consistent if } \Gamma \vdash \bot$$

We also need to introduce the notion of *maximally consistent sets.*

Definition: Characterization of inconsistent Set

$$\Gamma \vdash \bot \iff \forall \varphi . \Gamma \vdash \varphi \iff \exists \varphi . \Gamma \vdash \varphi \wedge \Gamma \vdash \neg \varphi$$

Definition: Maximally consistent set

$$\Gamma \nvdash \bot \Gamma \text{ is maximally consistent if } \ \Delta \nvdash \bot, \Gamma \subseteq \Delta \Longrightarrow \Gamma = \Delta$$

Let's define these other propositions:

Definition: Proposition 1

$$\Gamma \subseteq WWF_{\mathcal{L}} \text{ if } \Gamma \vdash \bot \text{ then } \forall L \text{ interpration}$$
$$U \wedge \rho_U, \rho_U \vDash \Gamma$$

If theres a structure that satisfy Gamma, the set is consistent. Equivalently:

$$\text{if } \exists U . \rho_U s.t. U . \rho_U \vDash \Gamma \Longrightarrow \Gamma \nvdash \bot$$

Our main proposition GOAL is:

$$\Gamma \subseteq (WWF)_{\mathcal{L}} \text{ if } \Gamma \nvdash \text{then } \Gamma \text{ sat.}$$

This is the key result for the completeness theorem.

3.1.1 Proof

We'll include only a fragment of the possible connectives: $\{\wedge, \rightarrow, \bot, \forall\}$. But we want to reason in an existential style since it's better to find a weakness in respect to find a universal argument. Let's define some abbreviations like:

$$\neg \varphi := \varphi \rightarrow \bot$$
$$\exists x . \varphi := \neg \forall x . \neg \varphi$$

The first defintion is the one of **Henkin Set**:

Definition: Henkin Set

$\Gamma \subseteq (WFF)_{\mathcal{L}}$ is an Henkin Set if for each $\exists x . \varphi(x)$ there exists a constant c such that:

$$\Gamma \vdash \exists x . \varphi(x) \rightarrow \varphi(c)$$

Notice that we extending the language.

Definition: Proposition 2

If Γ is an Henkin Set and $\Delta \subseteq WFF_{\mathcal{L}}$ with respect $\Gamma \subseteq \Delta$, Δ is an Henkin Set.

We'll now have to introduce the notion of "H-extension".

Definition: H-extension

This defintion is split in two parts:

- L language, we call "H-extension" of L a new language L^H obtained by adding to the set of constants of L the following set:

$$HC = \{c_\varphi \mid \exists x.\varphi \text{ is an L formula}\}$$

- Given $\Gamma \subseteq WFF_{\mathcal{L}}$, the "H-extension" of Γ with L^H formulas is:

$$\Gamma \cup \{\exists x.\varphi(x) \to \varphi(c_\varphi) \mid \exists x.\varphi \text{ is in } L \text{ and } c_\varphi \in HC\}$$

So I would like to connect derivability from the L language also in the H-extension.

Definition: Lemma 4

Suppose $\Gamma \cup \{\varphi\}$ and $\Gamma \vdash \varphi$. If I am deriving this in the H-extension of the language then I can derive it in the original language.

$$\Gamma \underset{L^H}{\vdash} \varphi \iff \Gamma \underset{L}{\vdash} \varphi$$

The cardinality fo the language:

$$\#[\mathcal{L}^{\mathcal{H}}] = \#[\mathcal{L}]$$
$$\#[\mathcal{L}] = \max(\#[R], \#[F], \#[C], \aleph_0)$$

Example

We know that $\aleph_0 = |\mathbb{N}|$. Also $|\mathbb{Z}| = |\mathbb{Q}| = \aleph_0$. Also if I take the set of $|\text{Prime}| = \aleph_0$. If you can find a bijection between the set of natural numbers and another set, you can say it's countable. Also $|\mathbb{R}| = 2^{\aleph_0}$.

We skip the proof of the Henkin extension (Lemma 5), but we can say that it is a very important result in order to prove the completeness theorem.

Definition: Lemma 6

Γ formulas is $WFF_{\mathcal{L}}$,

$$\forall \varphi \in \mathcal{L}, \quad \underbrace{\Gamma^H}_{\text{Henkin Extension}} \quad \vdash \varphi \Longrightarrow \Gamma \vdash \varphi$$

Another proposition that follows:

Definition: Proposizione 7

$\Gamma \subseteq WFF_{\mathcal{L}}$, if Γ consistent $(\Gamma \nvdash \bot) \Longrightarrow \Gamma^H \nvdash \bot$.

Definition: Henkin Closure

$\Gamma \subseteq WFF_{\mathcal{L}}$, consider the following sequence:

$$\Gamma_0 = \Gamma \text{ if } L = L_0$$

$$\Gamma_{i+1} = (\Gamma_i)^H \text{ if } L_{i+1} = L_i^H$$

The **Henkin closure** of Γ:

$$\Gamma^* = \cup_{\{i \in \mathbb{N}\}} \Gamma_i \qquad L^* = \cup_{\{i \in \mathbb{N}\}} L_i$$

Definition: Theorem 9

$\Gamma \subseteq WFF_{\mathcal{L}}$ and $\Gamma \nvdash \bot$:

1. $\#[\Gamma^*] = \#[\Gamma]$
2. $\Gamma^* \nvdash \bot$ (This proof is *required*)
3. Γ^* is a Henkin Set
4. $\forall \varphi \in WFF_{\mathcal{L}}$, $\Gamma^* \vdash \varphi \Rightarrow \Gamma \vdash \varphi$

If you want to a more classical proof, we need to use a very different semantics for the first order language:

Example

Proof of Point 2 of Theorem 9:

$$\Gamma^* \nvdash \bot$$

Suppose $\Gamma^* \vdash \bot$. Then there exists a finite set Σ^0 $(\Sigma^0 \subseteq \Gamma)$ such that $\Sigma^0 \vdash \bot$ and $\Sigma^0 \neq \varnothing$. $\Sigma^0 = \{\varphi_0, ..., \varphi_n\}$, there exist n such that:

$$\Gamma_{i_i}, ..., \Gamma_{i_n} \text{ in the Henkin sequence } \left(\text{not nec } \Gamma_{i_j} \neq \Gamma_{i_z}\right)$$

such that $\varphi_1 \in \Gamma_{i_1}$, dots, $\varphi_n \in \Gamma_{i_n}$. Let $m = \max(i_1, ..., i_n)$ then $\Sigma^0 \subseteq \Gamma_m$.

$$\Gamma_m \vdash \bot \Longrightarrow \text{impossible}$$

Definition: Intermezzo: Zorn's Lemma (Hausdroff's' 1914)

Let $< U, \leq >$ be a partially ordered set such that every $V \subseteq U$ which is totally ordered by $\leq$ has an upper bound in U. Then U has a maximal member z w.r.t $\leq$ (no $x \in U$ such that $x \geq z$).

Zorn's Lemma is equivalent to *Axiom of Choice*.

Definition: Theorem 10 (Lindenbaum's Lemma)

$\Gamma \subseteq WFF_{\mathcal{L}}$, $\exists \Gamma^{\text{MC}}$, $\Gamma \subseteq \Gamma^{\text{MC}}$ such that:

$$\Gamma^{\text{MC}} \text{ maximally consistent}$$

$$\Gamma^{\text{MC}} \text{ is an Henkin Set}$$

Example

Proof. Given that Γ^* is consistent and Henkin.

$$E(\Gamma^*) = \{\Delta \mid \Delta \nvdash \bot, \Delta \subseteq WFF_{\mathcal{L}}, \Gamma^* \subseteq \Delta\}$$

Take $< E(\Gamma^*), \subseteq >$, which is a partially ordered set. Let $\Sigma = \bigcup_k \Delta$ where k is a chain in $E(\Gamma^*)$.

- Σ is an upper bound of k.
- $\Sigma \nvdash \bot$.

For RAA, suppose $\Sigma \vdash \bot$. As for the previous proofs, there exists $\Sigma^0 \subset \Sigma$ such that $\Sigma^0 \vdash \bot$ which is not empty and I can say that $\Sigma^0 = \{\varphi_1, ..., \varphi_n\}$. then $\exists n$ elements $k_1, ..., k_n$ is the chain k such that $\varphi_1 \in \underbrace{k_1}_{\Delta_{i_j}}, ..., \varphi_n \in \underbrace{k_n}_{\Delta_{n_j}}$. Since K is an ordered chain $(\subseteq)$, $\exists K^+$ such that $\Sigma^0 \subseteq K^+ \Rightarrow K^+ \vdash \bot$. Which is not possible.

By Zorn's Lemma, if we consider the construction $< E(\Gamma^*), \subseteq >$ has an upper bound $(\Gamma^*)^{\text{MC}}$ which is an Henkin Set and also maximally consistent. So $\Gamma \subseteq \Gamma^*$. $\square$

In order to prove satisfiability, we need to define at least one model for the maximally consistent Henkin Set. We have to indroduce the notion of *syntactical modal*.

Definition: Proposition 11

Γ is maximally consistent:

1. Γ clsoed under derivability
2. $\forall \varphi \in WFF_{\mathcal{L}}, \varphi \in \Gamma$ or $\neg\varphi \in \Gamma$
3. $\forall \varphi. \neg\varphi \in \Gamma \iff \neg\varphi \notin \Gamma$
4. $\forall \varphi, \psi (\varphi \to \psi) \in \Gamma \iff \varphi \in \Gamma \to \psi \in \Gamma$

Definition: Proposition 12

Let Γ be a maximally consistent set. Let $\sim \subseteq WFF_{\mathcal{L}} x WFF_{\mathcal{L}}$ be $t \sim w$, $t, u \in \text{TERMS} \iff t = u \in \Gamma$:

$$\sim \text{ equivalent relation}$$

Also $\| t \|$ equivalent class of t:

$$\| t \| = \{u \in \text{TERMS} \mid t \sim u\}$$

$$\Gamma \setminus \sim \text{ is a quotient set}$$

Definition: Proposition 13

Γ maximally consistent set $t_1, ..., t_n, u_1, ..., u_n$ terms s.t. $\forall i \in [1, n] t_i = u_i$:

1. $\forall f$ function symbol:

$$f(t_1, ..., t_n) \sim f(u_1, ..., u_n)$$

2. $\forall R$ relational symbol

$$R(t_1, ..., t_n) \in \Gamma \implies R(u_1, ..., u_n) \in \Gamma$$

Definition: the syntactical model

$\Delta \in WFF_{\mathcal{L}}$ maximally consistent Henkin set.

$$\mathcal{U} =< \mathcal{C}^{\mathcal{L}} \setminus \sim, [.]^{\mathcal{U}} >$$

defined as:

1. $c^{\mathcal{U}} = \| c \|$, for the constant
2. $f \in \mathcal{F}^{\mathcal{L}}(n \text{ arity})$

$$f^{\mathcal{U}}(\|t_1\|, ..., \|t_n\|) = \|f(t_1, ..., t_n)\|$$

3. $R \in \mathcal{R}^{\mathcal{L}}$ then

$$R^{\mathcal{U}}(\|t_1\|, ..., \|t_n\|) \text{ iff } R(t_1, ..., t_n) \in \Gamma$$

This is also called the *syntactic interpratation.*

Definition: Theorem 15

$\Delta \in WFF_{\mathcal{L}}$ maximally consistent is the syntactic interpreation ρ_u is $\rho_{u(x)} = \|x\|$ then

1. $\mathcal{U}, \rho_{\mathcal{U}} \vDash \Delta$

(This one is not required but interesting)

1. U, rho_u is a model of Delta
   1. $\varphi \in \Delta \implies \mathcal{U}, \rho_{\mathcal{U}} \vDash \varphi$
   2. $\varphi \notin \Delta \implies \mathcal{U}, \rho_{\mathcal{U}} \nvDash \varphi$

   $\varphi := \forall x.\psi$:

a) $\forall x.\psi \in \Delta \overset{\text{Prop } 11 \wedge \forall E}{\Longrightarrow} \forall t.\psi\left[\frac{t}{x}\right] \in \Delta \overset{\text{IH}}{\Longleftrightarrow} \forall t\mathcal{U}, \rho_{\mathcal{U}} \vDash \psi\left[\frac{t}{x}\right] \Longleftrightarrow \forall t\mathcal{U}, \rho_{\mathcal{U}} \left[\!\left[|t^{\Downarrow_{\rho_{\mathcal{U}}}}_{x} \right]\!\right] \vDash \psi \Longleftrightarrow \forall t.\mathcal{U}, \rho_{\mathcal{U}} \left[\!\left[|t^{\Downarrow}_{x} \right]\!\right] \vDash \psi \Longleftrightarrow \mathcal{U}, \rho_{\mathcal{U}} \vDash \forall x.\psi \square$

Definition: Lemma 16

$\Gamma \subseteq WFF_{\mathcal{L}}$ s.t. $\Gamma \nvdash \bot$ then Γ is satisfafiable.

Example

Proof. Consider Γ^* (henkin set and MC) by TH15, we can define the syntactical model:

$$\exists \rho_{\mathcal{U}} s.t. \mathcal{U}, \rho_{\mathcal{U}} \vDash \Gamma^*$$
$$\Longrightarrow \mathcal{U}, \rho_{\mathcal{U}} \vDash \Gamma \qquad \square$$

Definition: Completeness Theorem

$$\Gamma \vDash \varphi \Rightarrow \Gamma \vdash \varphi$$

Example

Proof. Let's follow the equivalence $A \to B \equiv \neg B \to \neg A$ $\Gamma \nvdash \varphi$ then $\Gamma \nvDash \varphi$. If $\Gamma \nvdash \varphi \Rightarrow \Gamma, \neg\varphi \nvdash \bot$ then by Lemma 16, $\Gamma \cup \{\neg\varphi\}$.

$$\exists \mathcal{U}, \rho_{\mathcal{U}} \text{ s.t. } \mathcal{U}, \rho_{\mathcal{U}} \vDash \Gamma, \neg\varphi$$
$$\mathcal{U}, \rho_{\mathcal{U}} \vDash \Gamma \wedge \mathcal{U}, \rho_{\mathcal{U}} \nvDash \varphi$$
$$\text{then } \Gamma \nvDash \varphi \qquad \square$$

4 Intuionism

Little scheme for the next topics:

1. The Curry Howard Isomorphism
   1. Intuionistic Logic
      1. Intuionistic interpretation of connectives
      2. Classical principles ? $A \to B \equiv \neg A \vee B$, $\neg P \vee P$
      3. BHK semantics
      4. Kripke's semantics
      5. Drop ex falso and we will obtain Minimal Logic which is equivalent to simply typing language.
   2. Simply typed λ-calculus
   3. Normalization

$$\Pi_i \to \Pi_j \text{ compute with proofs}$$

$$\text{Formulas} \leftrightarrow \text{Types}$$

$$\text{Proofs} \leftrightarrow \text{Programs}$$

2. Proof theory of *Modal logic*

4.1 Constructivism

Connectives in intuitionistic logic are interpreted in a constructive way.

- $(\wedge)$: a proves $\varphi \wedge \psi := a$ is a pair $< b, c >$ such that b proves φ and c proves ψ.
- $(\vee)$: a proves $\varphi \vee \psi := a$ is a pair $< i, b >$ such that $i \in \{0, 1\}$ and if $i = 0$ then b proves φ and if $i = 1$ then b proves ψ.
- $(\to)$: a proves $\varphi \to \psi := a$ is a function that transforms any proof of φ into a proof of ψ.
- $(\bot)$: no a proves $\bot$. In order to deal with the quantifies we assume that some domain D of objects is given.
- $(\forall)$ a proves $\forall x \varphi(x) := a$ is a construction such that for each $b \in D, a(b)$ proves $\varphi(b)$.
- $(\exists)$ a proves $\exists x \varphi(x) := a$ is a pair $< b, c >$ such that $b \in D$ and c proves $\varphi(b)$.